

Encoder-only models

10

Encoder-Decoder models

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Decoder-only models

10

Early model training pipeline

1

Modern LLM families

Modern LLM training pipeline

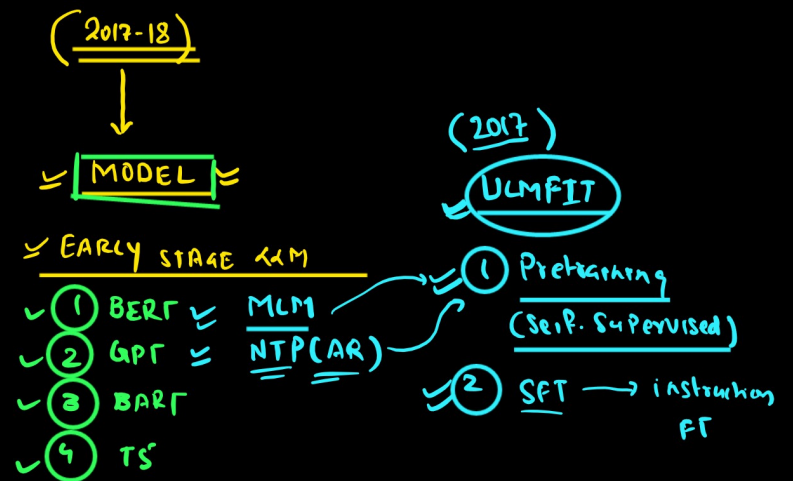
SLM concepts

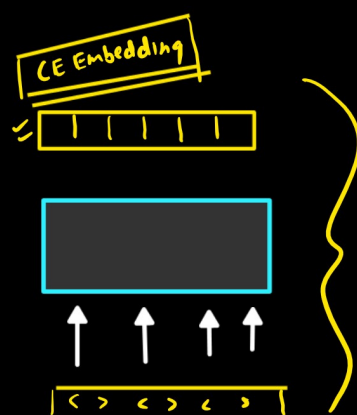
Multimodal concepts:

Multimodal model families

Leaderboards

APIs to access models







The heart of Transformer is self-attention, where the model understands each token in a sentence in the context of all the other tokens.

✓ Encoder-Only Models:

BERT ✓ ←

RoBERTa ✓

DistilBERT ✓

DeBERTa ✓

Use of these models:

Text classification: Sentiment analysis, Named Entity Recognition, POS

Embedding generation ✓

Input: "I hate this product."

Output: Negative

Input: "Virat Kohli lives in India."

Output: Virat Kohli = Person, India = Location

Input: "Ram eats mango."

Output: Ram = Noun, eats = Verb, mango = Noun

Encoder-only models are best for deeply understanding text, but they are not naturally designed to generate text.

Google / Meta / Apple

PAD / RS / AI

AI
Dev

90-95-99.1.

Encoder-Decoder Models

T5 ✓ → Google
BART ✓ → Meta
mT5 ✓ → Google
FLAN-T5 ✓ → Google

2019-2020

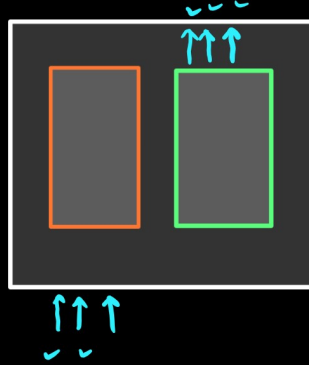
Use case:

Translation

Summarization

Question answering

T5/BART



Input: "How are you?"

Output: "आप कैसे हैं?"

Input: "Transformers use self-attention to understand relationships between words and are used in modern LLMs."

Output: "Transformers use self-attention and power modern LLMs."

Context: "BERT is an encoder-only model."

Question: "What type of model is BERT?"

Output: "Encoder-only model."

T5, BART

Encoder-decoder models are best when we need to understand the input and generate a new structured output, such as translation or summarization.

Decoder-Only Models:

GPT ✓
LLaMA ✓
Mistral ✓
Deepseek ✓

Use case: text generation

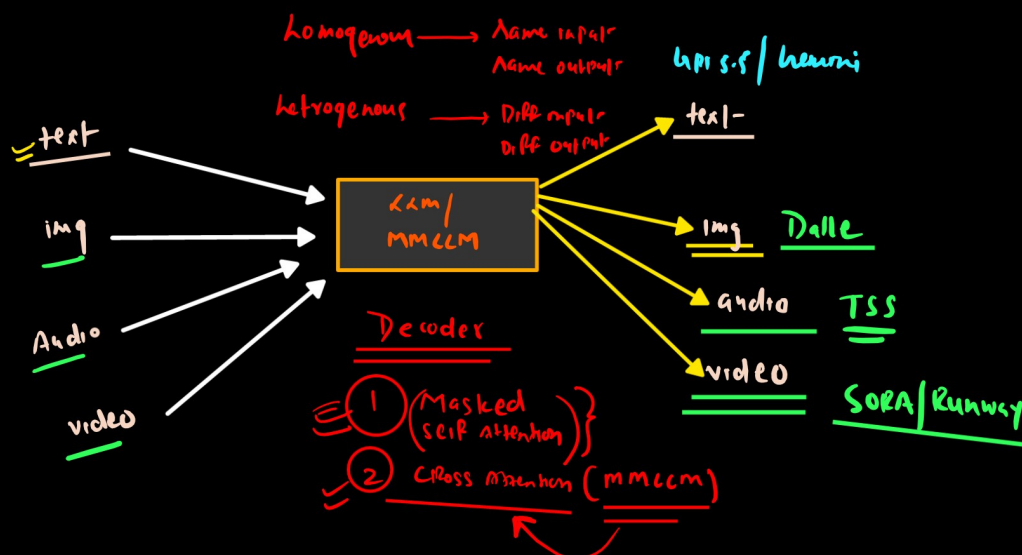
Decoder-only models are general-purpose generative models.

They predict the next token one by one and can generate many types of outputs.

It can generate any thing

- ✓ Conversational AI: Chatbots, QA assistants
- Code generation: Python, JavaScript, SQL, APIs
- Content writing: Blogs, emails, scripts, summaries
- Reasoning: Step-by-step problem solving

Today's most popular LLMs are based on the decoder-only style because their core task is next-token prediction and generation.



Early Stage Model Training

Self-supervised / unsupervised (pretraining)

Data → non-label data

Label is being created

Supervised fine-tuning

(BERT)

Text classification

Stage 1: Self-Supervised Pretraining

In this stage, the model learns language patterns from a large amount of unlabeled text data.

Example: BERT

BERT uses Masked Language Modeling.

(MLM)

Pretraining

Language Pattern

Input: The cat [MASK] on the mat.

Target: sat

Here, the model learns to predict the missing word using both left and right context.

Example: GPT

GPT uses Next Token Prediction.

Input: I love machine

Target: learning

Here, the model looks at previous tokens and predicts the next token.

Another example:

Input: Artificial intelligence is

Target: powerful

Then the sequence becomes:

Artificial intelligence is powerful

Now GPT predicts the next token again.

Input: Artificial intelligence is powerful

Target: because

So GPT learns by repeatedly predicting the next token from huge text data.

In pretraining, the model learns language structure, grammar, facts, and patterns without manually labeled data.

Stage 2: Supervised Fine-Tuning

In this stage, the model is trained on labeled data for a specific task.

Example:

Input: This movie is amazing.

Label: Positive

Another example:

Input: Translate "How are you?" into Hindi.

Output: आप कैसे हैं?

Teaching Line

Supervised fine-tuning converts a general pretrained model into a task-specific model using labeled examples.

Fine-tuning turns a general language model into a task-specialist model.

Modern LLM Training Pipeline

Correct modern pipeline:

Large-scale pretraining

↓

Supervised Fine-Tuning / Instruction Tuning

↓

Preference Alignment

Stage 1: Pretraining

The model is trained on internet-scale or curated data.

Goal:

Next token prediction

Learns:

Language Facts Reasoning patterns Code World knowledge Basic instruction patterns

Stage 2: Supervised Fine-Tuning / Instruction Tuning

The model is trained on human-written instruction-response pairs.

Example:

User: Explain Transformer in simple words. Assistant: Transformer is a neural network architecture...

Learns:

Instruction following Answer formatting Helpful response style Task completion

Stage 3: Preference Alignment

The model is aligned using human or AI preference data.

Methods:

RLHF DPO ORPO GRPO Constitutional AIRLAIF

Learns:

Helpful answer

Safe answer

Polite tone

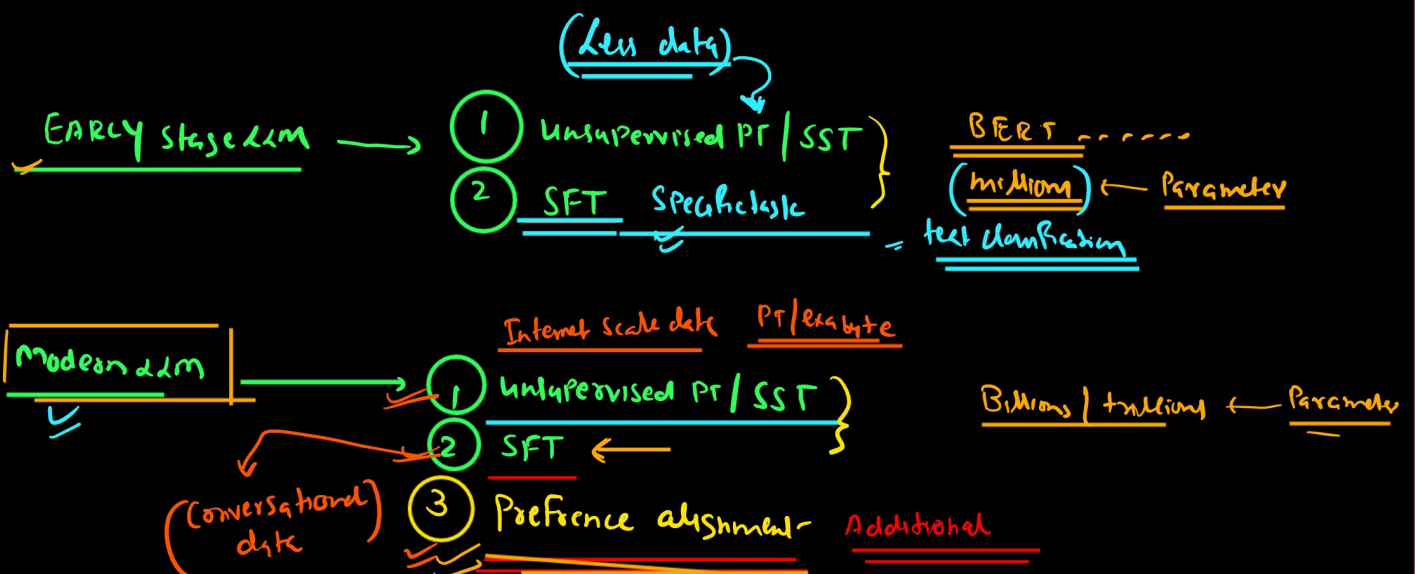
Better reasoning

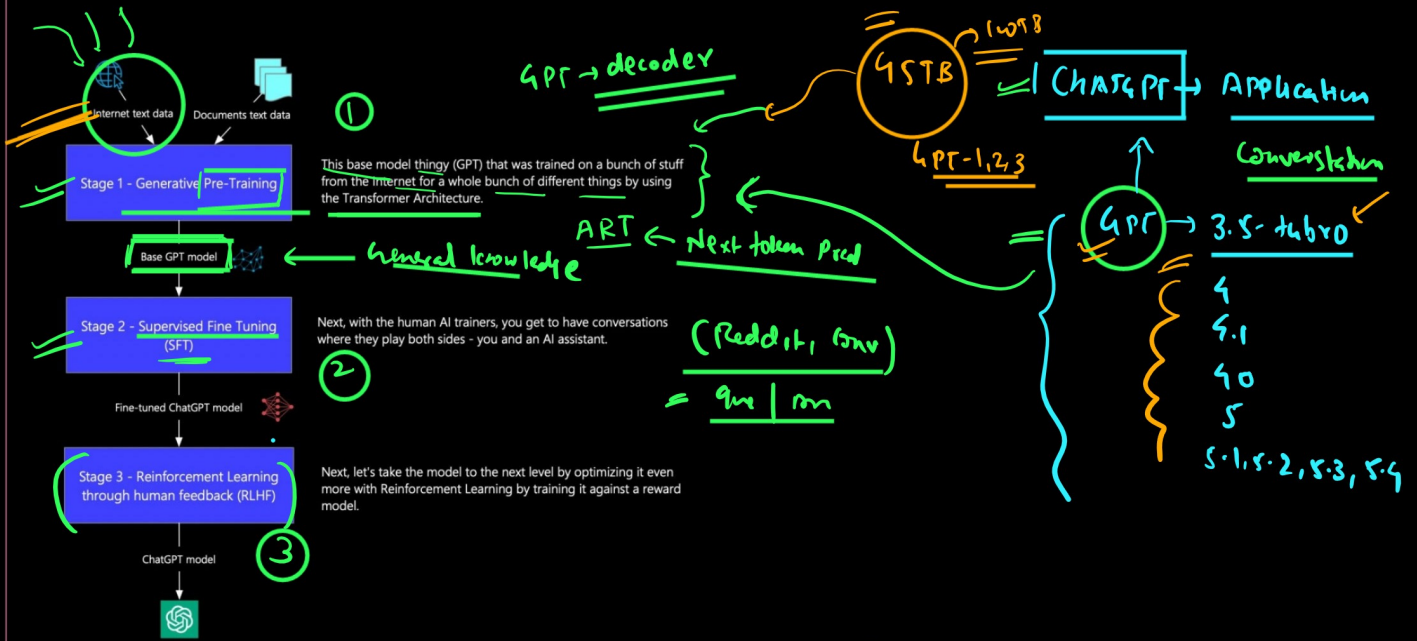
Less harmful output

Better refusal behavior

Teaching Line

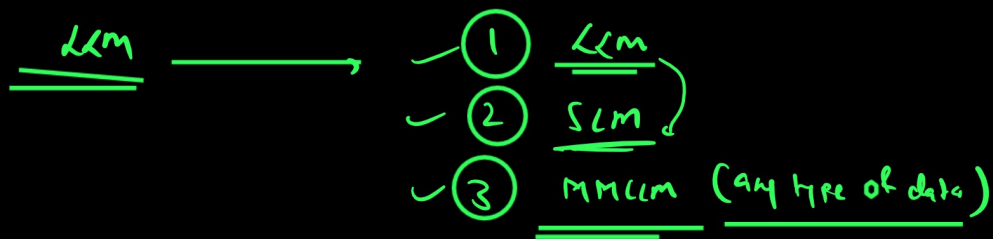
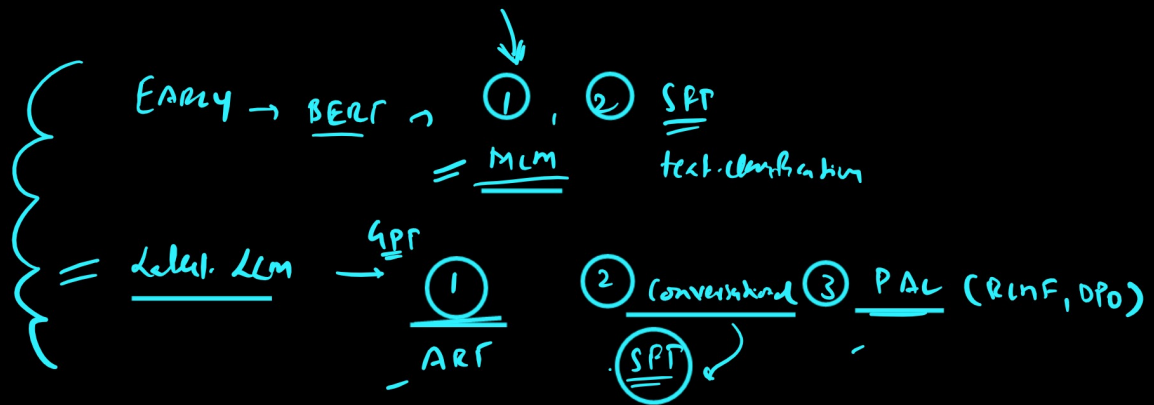
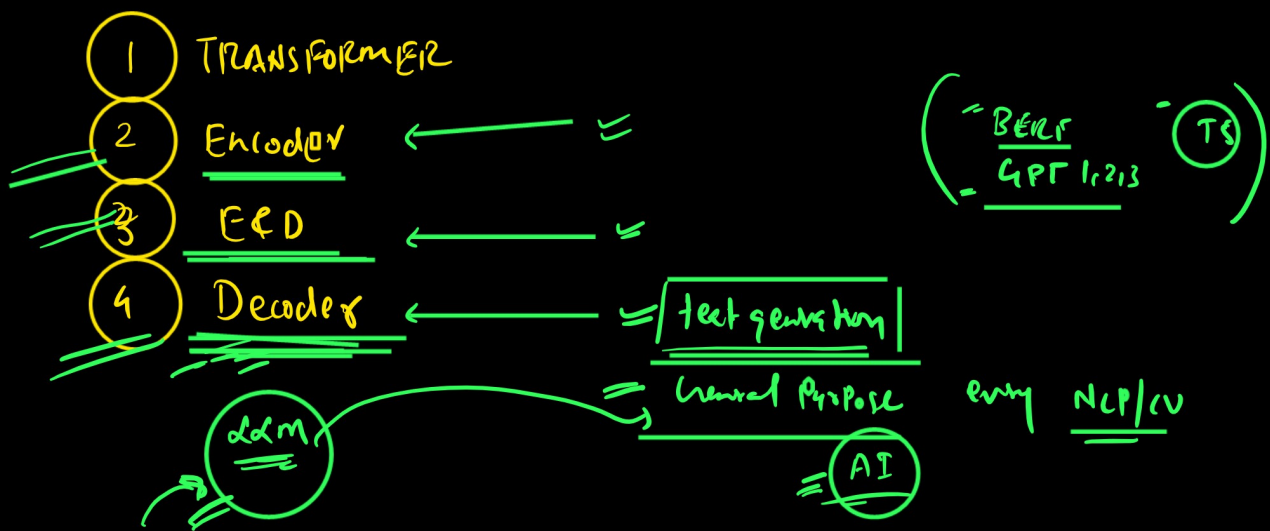
Pretraining gives the model knowledge, SFT teaches the model instruction following, and preference alignment makes the model behave like a human-like assistant.

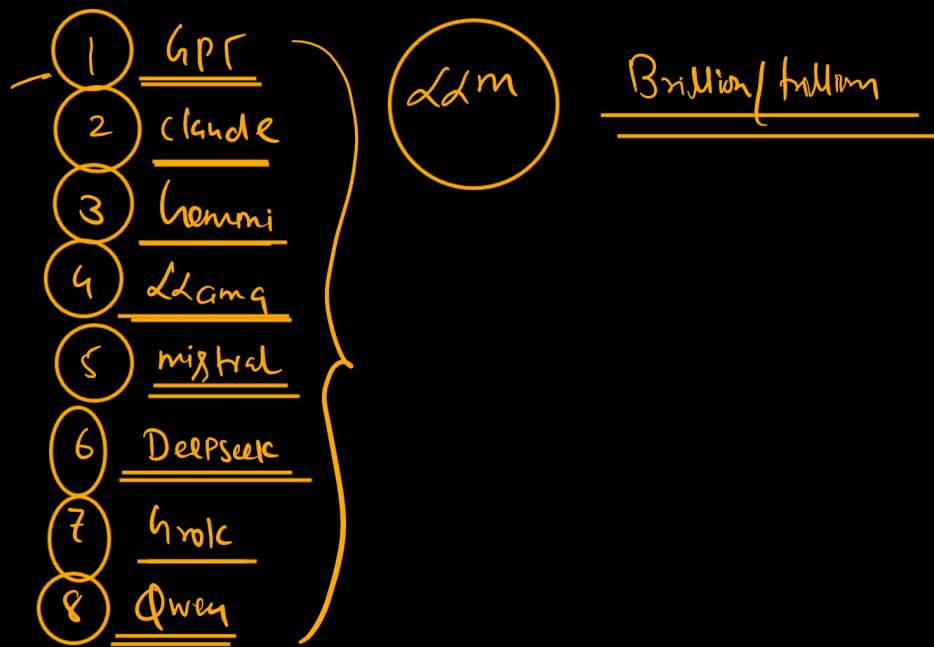




Question	Answer 1	Answer 2	Preference
How are you feeling today?	I am feeling happy	I am feeling intense	1
What did you think of the movie?	It was absolutely wonderful	It was surprisingly intense	1
How was your day at work?	I finished all my tasks	I struggled with deadlines	1
How do you view this project?	This project excites me	I have no opinion	1
Did you enjoy the meal?	The meal was bland	The meal was fantastic	2
How do you feel about traveling?	I love exploring new places	Traveling makes me nervous	1
What do you think of your team?	My team is very supportive	My team is sometimes difficult	1
How was your presentation?	I was nervous but confident	I was totally unprepared	1
Did you like the book?	The book was fascinating	The book was confusing	1
How did the interview go?	I think I made a good impression	I'm unsure how I did	1







model > job ← LLM

SLM Concepts

SLM = Small Language Model.

The goal of an SLM is:

Fewer parameters, lower cost, faster inference, and strong performance on specific tasks.

Less Parameters

Model < 10B

SLM

9B, 8B, 7B

Phi family

Gemma small variants

Qwen small variants

LLaMA small variants

Mistral 7B style models

Microsoft

Google

10B

SLM

SLM use cases:

On-device AI

Enterprise internal chatbot

Domain-specific RAG

Low-cost inference

Fast customer support bot

Private deployment

Edge AI

SLM

SARVAM AP

SLM

Date

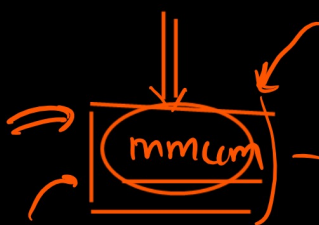
PT

12/13

Not every problem needs a GPT-level giant model. In many business use cases, a well-tuned SLM is cheaper, faster, and more controllable.

LLM/SLM

diff



GPT 5.5, Claude 4.2, Gemini 3

Audio
trans
text

{ text to text
text to image
img to text
img to img }

text

LLM

text

Image

Image

Vit, CLIP
Vision transformer

Stable diffusion

Images

Diffusion

Process

Denosing

Unet

transformer based denoiser

Multimodal Concept: ViT

ViT = Vision Transformer.

Concept:

Image → patches → patch embeddings → Transformer encoder → image representation

Example:

224x224 image

↓

16x16 patches

↓

Patch tokens

↓

Transformer

ViT

ViT treated images as patches, similar to words, and made Transformers powerful in computer vision.

The ViT paper showed that a pure Transformer can be applied directly to image patches and perform strongly on image recognition when trained at scale.

CLIP = Contrastive Language-Image Pretraining.

Concept:

Image encoder + Text encoder

Training:

Correct image-text pair ko close lao
Wrong image-text pair ko far karo

Example:

Image: dog running on grass
Text: "a dog running on grass"

CLIP image aur text ko same embedding space me map karta hai.

Use cases:

Image search
Zero-shot classification
Image-text matching
Text-guided image retrieval
Multimodal retrieval

CLIP brought images and text into a shared semantic space. It became a foundational block of multimodal AI.

1. ViT

→ Image ko patches/tokens me convert karta hai

2. CLIP

→ Image and text ko same semantic space me lata hai

3. BLIP / BLIP-2

→ Image understanding + captioning + VQA

4. LLaVA

→ Image ko LLM ke saath connect karke visual instruction following karta hai

5. Diffusion

→ Noise se image generate karna

6. Stable Diffusion

→ Text prompt se latent diffusion ke through image generate karna

7. VAE

→ Image ko compressed latent space me le jana

8. U-Net / DiT(Diffusion Transformer)

→ Denoising engine

9. Cross-Attention

→ Text prompt ko image generation guide karna

Diffusion = generation process

Transformer = architecture

Random noise → step-by-step denoise → final image/video/audio

Stable Diffusion 1.x

Stable Diffusion 2.x

SDXL

Inme main denoising network mostly:

U-Net

hota hai.

Flow:

Text prompt



Text encoder, often CLIP/T5



U-Net denoiser



VAE decoder



Final image

Prompt: "a dog wearing sunglasses"



CLIP Text Encoder



Text embedding



Diffusion U-Net / Transformer denoiser



Generated image

ViT ≠ Diffusion

CLIP ≠ Diffusion

DiT = Diffusion process + Transformer architecture

Diffusion model by default Transformer nahi hota.

Purane diffusion models mostly U-Net use karte the.

Modern diffusion models me Transformer-based denoiser use hota hai, jise DiT bolte hain.

Different Multimodal Model Names

Image-to-Text / Vision-Language Understanding

BLIP
BLIP-2
LLaVA
Flamingo
Kosmos
GPT-4V / GPT-4o style
Gemini Vision style
Qwen-VL

Text-to-Image Models

DALL-E
Stable Diffusion
SDXL
Imagen
Midjourney
FLUX
Firefly

Audio-to-Text Models

Whisper
Wav2Vec 2.0
HuBERT
DeepSpeech
Conformer

Text-to-Audio / Text-to-Speech Models

Tacotron 2
FastSpeech
VITS
Bark
AudioLM
MusicLM
ElevenLabs-style systems

Text-to-Video Models

Sora

Veo

Kling

Runway Gen series

Pika

Luma Dream Machine

Stable Video Diffusion

Video-to-Text / Video Understanding Models

VideoBERT

VideoCLIP

Video-LLaVA

VideoChatGPT

Gemini multimodal

GPT-4o style systems

Image-to-image models take an existing image as input and generate a modified image based on a prompt, style, mask, or control signal.

Input image + instruction/prompt



Modified / enhanced / transformed image

Image-to-Image Use Cases

Image editing

Style transfer

Inpainting

Outpainting

Background change

Sketch-to-image

Low-quality to high-quality image

Colorization

Pose/control-based generation

Object removal/addition

Popular Image-to-Image Models

Stable Diffusion img2img

ControlNet

InstructPix2Pix

SDEdit

DALL·E Editing

Photoshop Generative Fill

FLUX Kontext

IP-Adapter

InstantID

PhotoMaker

Input image: A normal dog photo

Prompt: Convert this dog into a cartoon style

Output: Cartoon-style dog image

Audio-to-audio: Audio-to-audio models take an audio input and generate a modified audio output, such as translated speech, converted voice, cleaned speech, or separated music.

Input audio



Modified / converted / enhanced audio

Audio-to-Audio Use Cases

Voice conversion
Speech-to-speech translation
Voice cloning
Noise removal
Audio enhancement
Music separation
Speaker style transfer
Audio super-resolution
Accent conversion
Dubbing

Popular Audio-to-Audio Model/System Examples

SeamlessM4T
AudioPaLM
RVC
Demucs
VoiceFixer
SpeechT5
OpenVoice
FreeVC
so-vits-svc

Simple Examples

1. Voice Conversion

Input: Your voice saying "Hello"

Output: Same sentence in another person's voice style

2. Speech-to-Speech Translation

Input: Hindi speech

Output: English speech

3. Noise Removal

Input: Noisy meeting recording

Output: Clean speech audio

4. Music Separation

Input: Full song

Output: Vocals + drums + bass + instruments separated

Video-to-video : Video-to-video models take an existing video as input and generate a modified video based on a prompt, style, mask, motion, or control signal.

Input video



Modified / transformed / enhanced video

Video-to-Video Use Cases

Video editing

Video style transfer

Background change

Object removal/addition

Face animation

Motion transfer

Video enhancement

Low-quality to high-quality video

Frame interpolation

Color grading

Text-guided video editing

Popular Video-to-Video Models / Systems

Runway Gen series

Pika

Luma Dream Machine

Sora-style video editing

Stable Video Diffusion

AnimateDiff

VideoCrafter

TokenFlow

Tune-A-Video

Rerender-A-Video

EbSynth

FILM / RIFE

Simple Examples

1. Video Style Transfer

Input: Normal walking video

Prompt: Convert it into anime style

Output: Same motion, anime-style video

2. Background Change

Input: Person talking in a room

Prompt: Change background to a studio setup

Output: Same person, studio background

3. Object Removal

Input: Road video with unwanted car

Prompt: Remove the car

Output: Clean road video

4. Video Enhancement

Input: Low-quality old video

Output: High-resolution enhanced video

Text/LLM APIs

OpenAI API
Anthropic API
Google Gemini API
Google Vertex AI
AWS Bedrock
Azure Cloud Foundry
Together AI
Groq
Open Router
Fireworks AI
Hugging Face Inference Providers
Replicate

Image APIs

OpenAI
Stability AI
Replicate
Hugging Face
Fal.ai
Runware
Together
Fireworks

Audio APIs

OpenAI
ElevenLabs
Deepgram
AssemblyAI
Google Speech
Azure Speech

Video APIs

Runway
Pika
Luma
Replicate
Fal.ai
Google Veo access paths
OpenAI Sora access paths

